

Erdős Problem #662

EP-662

L1 - TRACTABLE

PARTIALLY SOLVED

Geometry

ORIGINAL PROBLEM STATEMENT

Consider the triangular lattice with minimal distance between two points 1. Denote by $f(t)$ the number of distances from any points $\leq t$. For example $f(1) = 6$, $f(\sqrt{3}) = 12$, and $f(3) = 18$.

Let $x_1, \dots, x_n \in \mathbb{R}^2$ be such that $d(x_i, x_j) \geq 1$ for all $i \neq j$. Is it true that, provided n is sufficiently large depending on t , the number of distances $d(x_i, x_j) \leq t$ is less than or equal to $f(t)$ with equality perhaps only for the triangular lattice?

In particular, is it true that the number of distances $\leq \sqrt{3} - \epsilon$ is less than 1?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>